Flagging Rules Without Admin

Every rule below is run by the system. No person has to decide anything.

> **Honest note.** Research backs the *ideas* (small steps first, two people must agree, explain every penalty). It does not give the *numbers*. All points, limits and time windows are starting settings for the pilot. Tune them with real data. Rules marked **New** are not built yet.

1. How it works (5 steps)

1. Something happens, or a user sends a report.
2. The system checks that it was the user's own fault.
3. If it counts, points are added to the user's score.
4. The score decides the level (Table 3).
5. The user gets a short notice: what happened, which rule, how to avoid it.

A report alone is **0 points**. It is only a claim until a second, different person agrees.

2. Violations and weights

Table 1. Buyer

Violation	How the system knows	Points	Status
Reservation ended, no pickup	Timer runs out	2	Built
Cancel late (under 3 hours before pickup)	Cancel time vs pickup time	1	New (code counts cancels now)
Decline at pickup, reason not on the list	Reason picked in app	1	New
Rude or abusive messages	2 different people report it	2	Built
QR misuse	2 different people report it	3 and block now	Built
Fake reports on sellers	3 or more reports by one user, none confirmed in 14 days	1 for each group of 3	New
Review bomb (see 4.2)	Burst of low ratings from new accounts	0 for the buyers, ratings ignored	New

Table 2. Seller

Violation	How the system knows	Points	Status
Cancel a confirmed order	Order record	2	Built
No reply to a buyer in 48 hours	Message times	1	Built
Low rating (1 or 2 stars)	Buyer wrote a reason, or it is the 2nd low rating in 30 days	1	Partly (code ignores the reason rule)

Violation	How the system knows	Points	Status
Wrong listing (weight, type, size)	2 different people report it	2	Built
Rude or abusive messages	2 different people report it	2	Built
QR misuse	2 different people report it	3 and block now	Built
Bad produce, buyer declined with a good reason	Seller accepts, or stays silent for 24 hours	2	New
Orders from own account	Buyer account is the seller account	3 and block now	New
Same buyer floods one seller with orders	More than 5 orders in 7 days from one buyer	0, extra orders do not count for ranking	New

Table 3. Both

Violation	How the system knows	Consequence	Status
Spam	Same message sent 5 or more times in 10 minutes	Cannot send for 10 minutes. 2nd time in 90 days: 1 point	New
Sharing personal data	Phone or email pattern in a chat message	Message is hidden. 3rd time in 90 days: 1 point	New

Stopped by design (no flag needed)

- **Price tricks:** the price is locked when the order is made.
- **Listing with no stock:** orders stop at 0 stock.

Never counts (0 points)

One report alone. The other side caused it (seller closed the stall). An excused case. An event that was fixed after a check (section 5).

3. Levels

Same for buyers and sellers, so it is simple and fair.

Score (S)	Level	Buyer	Seller
0 to 1	Good standing	Nothing	Nothing
2 to 3	Notice	Plain notice	Plain notice
4 to 5	Warning	Warning shown at reservation	Warning shown on dashboard
6 to 7	Pause 24 hours	No new reservations	No new listings, no Priority Boost
8 to 11	Pause 3 days	No new reservations	No new listings
12 or more	Block	Account blocked	Account blocked

- A penalty starts only when a new event moves the user **up** a level. If the score falls, the level falls with no new penalty.
- **Seller gate:** a seller is paused or blocked only if the fault rate F is 10% or more and they have at least 10 orders. Otherwise they stay at Warning and are marked "Under watch".

- **Serious cases** (QR misuse, orders from own account) skip the steps and block at once.

4. Formulas

4.1 Trust score

> **S = sum of points of all counted events in the last 90 days**

Old events drop off after 90 days, so a user can recover by behaving well. A Block lifts on its own when S falls under 12.

4.2 Review bomb check

> **A rating is ignored if:** its account is under 7 days old, **and** 5 or more such ratings of 1 or 2 stars hit one seller within 24 hours.

Ignored ratings do not count for the seller's score or average. The buyers get 0 points and a notice, because some may be honest.

4.3 Seller fault rate

> **F = (seller cancels + confirmed wrong listings + accepted declines) ÷ (completed + cancelled + declined orders)**, over 90 days, only with 10 or more orders.

A big seller with 3 faults in 500 orders has F = 0.6%, so no pause. A small seller with 4 faults in 10 orders has F = 40%, so the pause applies. Same score, fair result for both sizes.

5. Fixing a wrong penalty (no admin)

- **Excuse (New):** after an expired reservation, the buyer has 24 hours to press "I had a problem". The seller confirms. If yes, 0 points. Once every 30 days.
- **Check again (partly built):** within 7 days a user can ask for a re-check. The system re-reads the order record. If the record does not support the penalty, the event is removed and S is recalculated. Cancellation and no-reply events already work this way.
- **Decline at pickup, seller disagrees (New):** nobody gets points and the order closes as "Declined at pickup". The produce is not put back in stock.
- **Away mode (New):** the 48-hour reply clock stops while the seller is away. Up to 14 days each time.

6. Example

Ana (buyer). Day 1 reservation ends: S = 2, Notice. Day 10 cancels 2 hours before pickup: S = 3. Day 20 reservation ends: S = 5, Warning. Day 25 seller closes the stall: 0 points. Day 38 cancels 1 hour before: S = 6, Pause 24 hours. Day 70 after more events S = 12, Block.

7. Why each rule is fair (and the source)

Rule	Why	Source	Strength
Small penalty first, bigger later	Banning at once breeds anger. Steps teach.	Ostrom 1990 (graduated sanctions)	Strong idea
Add points into one score	One event should not decide a person.	Resnick et al. 2000	Medium
Report is 0 points, 2 people must agree	Flags are blunt and can be used to attack people. Feedback can be revenge.	Crawford and Gillespie 2016; Resnick and Zeckhauser 2002	Strong
Only count the user's own fault	Marketplaces count seller-fault events only.	Shopee penalty pages (company, reported by sellers' tools)	Weak: company practice
Fault rate, with a minimum order count	Big sellers have more faults just from more orders.	Shopee non-fulfilment rate (company)	Weak: company practice
Plain notice, re-check, excuse	People need to understand automatic decisions and get a chance to answer.	Citron 2008; Citron and Pasquale 2014	Strong for the principle
Cancel and no-show rules	Marketplaces limit habitual late cancels.	Too Good To Go partner terms (company)	Weak: company practice
Review bomb check	Fake reviews are common and come from weak or new accounts. Bursts and new accounts are standard warning signs.	Luca and Zervas 2016; Fei et al. 2013 (burst detection)	Medium. The 5 ratings, 24 hours and 7 days are my starting numbers
Orders from own account and order floods	Sellers place fake orders to climb rankings (called "brushing").	Jin, Yang and Hosanagar 2023	Medium. The 5 orders in 7 days is my starting number
Spam limit	Capping actions per time window is the usual defence.	Marketplace guidance (vendor blogs)	Weak: vendor advice
Hide phone and email in chat	Platforms mask contact details and payment details in messages.	eBay messaging rules; chat-moderation vendors	Weak: company practice
90-day window, all point values, level cut-offs	Team settings. No study gives them.	None	Team setting

8. References

Check each one against the original before you put it in your paper. I read search summaries only, not full papers.

1. Ostrom, E. (1990). *Governing the Commons.* Cambridge University Press.
2. Resnick, P., Kuwabara, K., Zeckhauser, R., and Friedman, E. (2000). Reputation systems. *Communications of the ACM, 43*(12), 45 to 48.
3. Resnick, P., and Zeckhauser, R. (2002). Trust among strangers in internet transactions: Empirical analysis of eBay's reputation system. *Advances in Applied Microeconomics, 11*, 127 to 157.
4. Crawford, K., and Gillespie, T. (2016). What is a flag for? Social media reporting tools and the vocabulary of complaint. *New Media and Society, 18*(3), 410 to 428.
5. Citron, D. K. (2008). Technological due process. *Washington University Law Review, 85*(6), 1249 to 1313. Citron, D. K., and Pasquale, F. (2014). The scored society. *Washington Law Review, 89*(1), 1 to 33.

6. Luca, M., and Zervas, G. (2016). Fake it till you make it: Reputation, competition, and Yelp review fraud. *Management Science, 62*(12), 3412 to 3427. Record
7. Jin, Y., Yang, S., and Hosanagar, K. (2023). To brush or not to brush: Product rankings, consumer search, and fake orders. *Information Systems Research, 34*(2), 532 to 552. Link
8. Fei, G., et al. (2013). Exploiting burstiness in reviews for review spammer detection. ICWSM. PDF. *Author list and year not confirmed. Verify.*
9. Company guidance, not research: Shopee penalty and non-fulfilment pages (Ginee, BigSeller); Too Good To Go Local Hero; eBay sharing contact details; chat moderation guide.

9. Not built yet

New rules need new code: late-cancel points, decline at pickup, fake-report group, review-bomb check, self-order and order-flood checks, spam limit, personal-data hiding, excuse, Away mode. The built rows already match `backend/src/utils/trust/policy.ts`. Level lengths differ today (12, 24 and 72 hours at 6, 8 and 10 points). Pick one table.